

Andrew Z. Ho

SOFTWARE ENGINEER · FULL-STACK · CLOUD & DISTRIBUTED SYSTEMS

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EXPERIENCE

Capital Group

Jun 2025 – Present

Software Development Engineer

TypeScript, Node.js, AWS, Vue 3, Terraform

- Owned an opt-in **Daily Digest** end-to-end — DynamoDB schema, a scheduled **AWS Lambda** aggregator, preferences API, and Vue UI — collapsing **50+** daily investor alerts into one market-close email; the platform's first digest type.
- Spearheaded **producer self-service editing**: a Fastify PUT API with immutable-field validation plus a Vue admin-console overhaul (template/Adaptive-Card editor, live preview, env routing), removing engineering from routine changes.
- Re-platformed the notification client from a **v1** → **v2 REST API** behind a flag and built backend **cursor pagination** (limit/nextToken) with infinite scroll, keeping latency flat as histories grew.
- Hardened a concurrent lock-based send pipeline with **typed exceptions**, proper **HTTP 409** on duplicate sends, and **Datadog** monitors — cutting on-call noise and preventing double-delivery.
- Shipped **~100 PRs across 7 repos** (TypeScript backend, Vue Module-Federation micro-frontends, Terraform IaC) and drove **3 epics** end-to-end in ~11 months.

New York Proton Center

Jun 2024 – Mar 2026

Software Engineering Consultant

Python, SQL · Remote · Sep 2024 – Mar 2026

- Built a proton-therapy **WET calculator** and Python/Tkinter visualizer on the **RayStation DICOM API**, cutting manual treatment-planning time by **25%**.

Software Engineering Intern

Snowflake, SQL, Python, Power BI · New York, NY · Jun 2024 – Sep 2024

- Architected a **Snowflake** star-schema warehouse unifying financial/operational data, accelerating month-end close by **30%**.
- Built incremental **SQL ETL pipelines** that cut the data-freshness SLA from **24h to 4h**.
- Shipped **10+ Power BI dashboards** with finance leads, enabling **20%** faster budget approvals.

Convo

Jan 2025 – Jul 2025

Software Engineer

Python, Flask, Socket.IO, WebRTC · Remote

- Designed **25+ RESTful Flask endpoints** with validation and error handling, cutting bug-related tickets by **30%**.
- Built a **Socket.IO/WebSocket** service for real-time transcription updates, lifting engagement by **20%**.
- Engineered a **JWT auth system** (token rotation, rate limiting, self-service reset), cutting unauthorized-access attempts **40%** and lockout downtime **50%**.

SKILLS

Languages	TypeScript, JavaScript, Python, SQL, C++, C, C#, Java, HCL/Terraform, HTML, CSS
Backend & Cloud	Node.js, AWS (Lambda, DynamoDB, SES/SNS/SQS, ECS Fargate, API Gateway, EventBridge, Secrets Manager, ElastiCache/Redis), Fastify, FastAPI, Flask, PostgreSQL, MongoDB, Snowflake
Frontend	React, Next.js, Vue 3 (Composition API), Webpack Module Federation / micro-frontends, Tailwind
Tools & Practices	Docker, Git, CI/CD, Terraform, Datadog, LaunchDarkly, Okta JWT/OAuth, pytest, Power BI, Agile/Jira, event-driven architecture, system design
AI & Data	Claude Code, GitHub Copilot, Cursor, OpenAI API; ML, NLP, ETL pipelines, AI model deployment

EDUCATION

University of California, Irvine

Mar 2025

Bachelor of Science, Computer Science

Irvine, CA